

HUTCHINSON, MN, USA

WMO: 726569

Lat: 44.859N

Lon: 94.382W

Elev: 1060

StdP: 14.14

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 04933

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-14.7	-9.0	-21.5	1.8	-13.8	-16.3	2.4	-9.0	28.3	11.3	25.9	13.9	6.0	300	0.565

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.8	90.0	75.0	86.4	72.7	83.3	71.0	78.0	85.8	75.7	82.8	73.6	80.3	9.8	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
75.4	138.4	83.1	73.2	128.5	80.2	71.7	121.9	78.5	42.3	85.6	39.9	82.9	37.9	80.0	86.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
24.7	20.9	18.7	-20.4	95.0	6.8	2.8	-25.2	97.0	-29.2	98.7	-33.0	100.3	-37.9	102.3		
			-20.7	81.7	6.3	2.2	-25.3	83.3	-29.0	84.6	-32.6	85.8	-37.2	87.4		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	44.8	14.2	17.7	30.3	45.1	57.3	67.7	71.7	68.7	61.7	48.0	33.4	20.4
	DBStd	22.49	13.52	13.05	12.57	9.81	8.74	6.70	5.85	5.38	8.36	9.31	10.48	11.95
	HDD50	4451	1111	906	620	203	29	0	0	0	9	151	505	917
	HDD65	7951	1576	1326	1077	598	270	46	11	24	160	533	948	1382
	CDD50	2559	0	0	8	56	256	532	671	580	360	89	7	0
	CDD65	584	0	0	0	2	32	127	217	139	61	6	0	0
	CDH74	5281	0	0	2	41	381	1142	1974	1114	551	74	2	0
	CDH80	1531	0	0	0	10	125	356	627	257	142	14	0	0
	WSAvg	8.1	8.9	8.8	9.3	10.2	9.4	7.7	6.0	5.5	6.6	8.2	8.9	8.4
	PrecAvg	30.0	0.6	0.6	1.5	2.7	3.7	4.8	3.8	4.1	3.1	2.3	1.8	0.7
Precipitation	PrecMax	39.2	1.5	1.1	3.9	7.0	6.9	9.1	8.1	7.6	7.0	5.9	5.1	2.1
	PrecMin	20.8	0.0	0.0	0.2	0.5	1.1	2.5	1.3	0.4	0.4	0.4	0.1	0.2
	PrecStd	4.9	0.4	0.4	0.9	1.4	1.5	1.7	1.8	1.8	1.8	1.5	1.5	0.5
	PrecStd	4.9	0.4	0.4	0.9	1.4	1.5	1.7	1.8	1.8	1.8	1.5	1.5	0.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	43.3	50.8	68.0	79.3	90.2	93.0	94.5	90.3	89.8	81.6	66.2	51.4
		MCWB	38.2	46.9	57.4	60.2	67.8	73.7	79.8	77.4	73.1	63.2	55.1	46.0
	2%	DB	37.3	42.3	59.1	72.4	82.8	88.1	90.1	85.9	83.8	74.6	58.9	43.3
		MCWB	35.0	38.4	53.5	55.8	63.4	71.4	77.5	75.5	70.0	61.0	49.8	38.6
	5%	DB	34.5	37.5	53.6	67.0	77.8	83.9	86.5	82.3	80.7	69.2	53.5	38.4
		MCWB	33.1	35.4	49.5	53.7	62.2	68.9	75.0	72.1	67.5	58.6	47.0	35.7
	10%	DB	32.3	35.5	46.5	61.4	72.9	80.8	82.9	80.6	76.7	63.2	48.1	35.7
		MCWB	31.3	33.9	42.5	50.0	59.6	67.0	72.4	70.4	66.1	54.6	42.6	33.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	38.6	47.3	61.2	62.9	72.0	77.8	82.3	79.6	76.4	68.3	58.3	46.8
		MCDB	42.3	50.1	64.4	74.0	84.2	87.3	90.3	87.0	84.0	75.3	64.8	48.9
	2%	WB	35.3	38.9	53.8	58.5	68.2	74.8	79.3	76.8	73.1	63.3	51.3	38.8
		MCDB	37.5	41.5	57.7	68.6	76.6	83.2	87.3	83.7	79.5	70.8	57.9	42.3
	5%	WB	33.2	36.0	48.2	55.0	65.2	71.9	76.8	74.4	70.7	59.4	46.6	36.0
		MCDB	34.3	37.8	53.2	63.9	73.3	80.2	84.1	80.5	77.3	66.7	52.0	38.5
	10%	WB	31.2	33.6	42.6	51.8	62.1	69.6	74.4	72.3	67.9	55.6	43.0	33.6
		MCDB	32.2	34.9	46.8	60.5	70.0	77.1	81.3	78.0	74.6	62.3	48.1	35.0
Mean Daily Temperature Range	5% DB	MDBR	16.1	17.5	16.4	19.9	20.7	19.5	19.8	20.0	22.1	20.1	16.7	15.3
		MCDBR	16.3	18.0	24.0	29.9	29.1	24.7	21.6	22.1	25.9	28.7	25.6	20.1
		MCWBR	14.7	15.0	17.0	17.1	14.2	11.5	11.9	12.6	13.6	16.0	17.3	15.9
	5% WB	MCDBR	15.9	16.4	23.2	25.8	23.6	20.1	20.0	19.9	21.6	23.7	22.3	18.4
		MCWBR	14.2	13.8	17.0	16.5	13.8	11.6	12.0	12.6	13.2	14.8	16.1	14.9
Clear-Sky Solar Irradiance	taub		0.258	0.280	0.307	0.361	0.389	0.394	0.399	0.405	0.368	0.317	0.283	0.260
	taud		2.456	2.388	2.434	2.341	2.318	2.354	2.341	2.307	2.383	2.525	2.527	2.464
	Ebn at Noon		279	290	296	285	279	278	275	268	270	273	267	266
	Edh at Noon		22	28	31	37	40	38	38	38	33	25	21	20
All-Sky Solar Radiation	RadAvg		564	899	1191	1472	1705	1900	2042	1693	1321	852	548	407
	RadStd		66	102	115	141	134	145	74	132	97	115	52	45

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (66 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page